

OPEN BY DESIGN **Open Science**



Open infrastructure for equitable cooperation

How Brazilian public infrastructure can advance a shared Global South Open Science agenda.

Washington Segundo

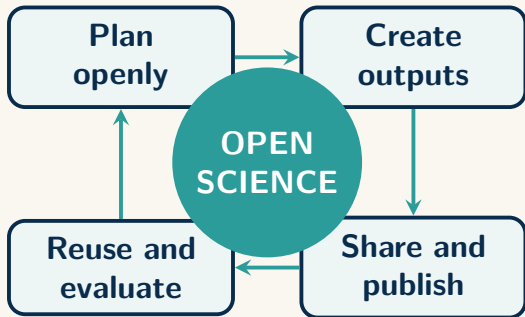
General Coordinator for Scientific and Technical Information

Brazilian Institute of Information in Science and Technology

Ministry of Science, Technology and Innovation



Open science redesigns the entire research cycle

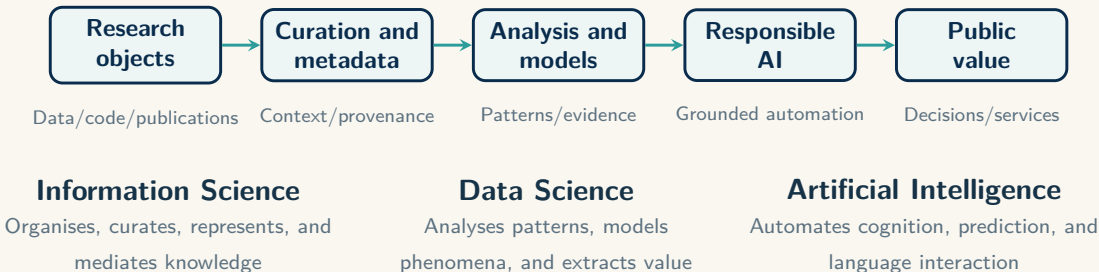


A PARADIGM SHIFT

- Openness begins at planning, not after publication.
- Publications, data, code, methods, and reviews become reusable outputs.
- Governance, preservation, and participation are part of scientific quality.

Core idea: knowledge should be accessible, auditable, interoperable, and useful to society.

Information Science is the foundation of reliable AI



A long evolution led from private exchange to open ecosystems



Digital networks separated the traditional functions of journals: record, certification, dissemination, and preservation.

Ibict has shaped scientific information

1954

Founded as Brazil's national
institute dedicated to
bibliography and scientific
documentation

Today

A research unit of the
Ministry of Science,
Technology and Innovation
- MCTI

A LONG-STANDING PUBLIC MISSION

- Organize, preserve, and disseminate Brazil's scientific and technological information.
- Build national services, standards, and interoperable information infrastructures.
- Connect Brazilian research with regional and international knowledge networks.

A STRONG TRADITION OF OPENNESS Ibict has played a sustained leadership role in **Open Access** and **Open Science**, advancing repositories, aggregators, research data, digital preservation, persistent identifiers, and public-interest governance.

Four axes turn information into public value

Information integrity and justice

Preserve authorship and meaning across the information cycle while ensuring equitable production, access, and use.

Information sovereignty

Strengthen public control over information flows, service infrastructures, algorithms, standards, software, and hardware.

Information for citizenship

Build critical, media, and data literacy for a multilingual, accessible, and socially inclusive information environment.

Open Science

Open research practices strengthen public trust, cooperation, and science diplomacy.

Information Science connects the four axes to support social, environmental, and economic development.

Source: Braga, *Ciência da Informação*, vol. 55, no. 1, 2026.

A Global South action plan turns principles into action

AFRICAN UNION · BRAZIL · CHINA · SOUTH AFRICA

Shared principles: equality and mutual benefit | respect for diversity
universal access | open collaboration

1 Global governance

Advance UNESCO principles through joint governance and monitoring.

2 Open data cooperation

Connect platforms for ethical and sustainable sharing.

3 Shared infrastructure

Co-develop public services, trusted data spaces, and AI for science.

4 Open scholarly communication

Share papers and preprints through inclusive infrastructures.

5 Public engagement

Build culture, education, communication, and participation.

6 Capacity in developing countries

Invest in skills, exchange, local capability, and open technology.

The ecosystem is broader than open access

Open outputs

Publications, research data, software, methods, and educational resources.

Open processes

Preprints, transparent peer review, responsible assessment, and reproducible workflows.

Open infrastructures

Repositories, identifiers, metadata standards, preservation, and research information systems.

Open participation

Citizen science, knowledge dialogue, inclusion, and public engagement.

Infrastructure connects every pillar and keeps openness sustainable.

Four principles balance technical value and social responsibility

FAIR

Make data findable, accessible, interoperable, and reusable for people and machines.

TRUST

Build transparent, responsible, user-focused, sustainable, and technically sound infrastructures.

CARE

Protect collective benefit, authority, responsibility, and ethics in data governance.

DEIA

Embed diversity, equity, inclusion, and accessibility in research systems.

Rule of thumb: as open as possible, as closed as necessary.

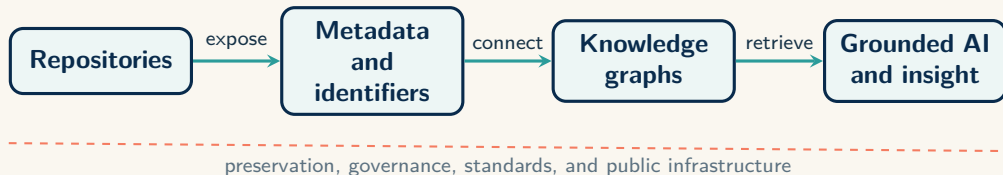
Maximize openness effectively, strategically, inclusively, and responsibly

Open publishing must close the scientific information divide

Route	How it opens access	Main strength	Main risk
Green	Deposit in a repository	Uses public infrastructure and can avoid publication fees	Embargoes or restrictive publisher policies
Gold	Immediate access on the publisher platform	The version of record is openly available	APCs can shift exclusion from readers to authors
Diamond	No fees for readers or authors	Aligns publishing with the public interest	Requires durable institutional funding and shared governance

Inclusive journals, repositories, and preprints must expand circulation without shifting exclusion from readers to authors.

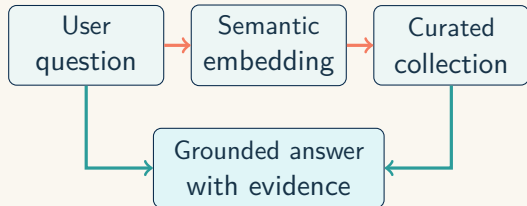
Trustworthy AI begins with trustworthy research infrastructure



Preservation and interoperability turn stored objects into reusable evidence — and enable the trusted data spaces and AI applications called for in Objective 3.

Semantic systems connect meaning, not only keywords

A GROUNDED RETRIEVAL FLOW



WHAT MAKES IT WORK

- Controlled vocabularies and ontologies reduce ambiguity.
- Multilingual embeddings retrieve concepts across languages.
- RAG grounds generated answers in real documents.
- Rich metadata makes sources traceable and reusable.

Applied AI creates value when the information base is ready

Application	AI capability	Public-interest value
ServicesPI	Semantic search + NLP	Natural-language discovery across complex patent collections
Patente.IA	Text + image similarity	Broader detection of related inventions and prior art
RDAPP	NLP + semantic retrieval	Structured access and analysis across public-policy documents
SAVITS	BI + ML + LLM	Monitoring, prediction, and explainable impact assessment

The recurring architecture is simple: **infrastructure** → **structured data** → **AI application**.

Openness must also deliver informational justice

THREE DIMENSIONS

- **Distribution:** equitable access to information and infrastructure.
- **Recognition:** respect for diverse people, knowledge, and epistemologies.
- **Participation:** meaningful inclusion in producing and using knowledge.

ETHICAL SAFEGUARDS

- Consent and risk assessment
- Privacy and controlled access
- Collective and Indigenous data rights
- Protection from harmful or dual use

Implementation depends on human mediation

INCLUSIVE DESIGN Build for diverse communities

- Accessible interfaces
- Multilingual discovery
- Disciplinary flexibility
- Context-sensitive workflows

GOVERNANCE Define responsibility

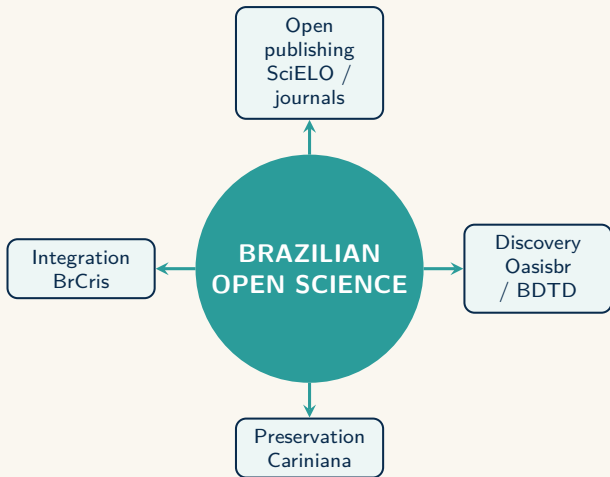
- Data control and privacy
- Curation criteria
- Ethical safeguards
- Sustainable ownership

DIGITAL LITERACY Enable critical use

- Practical training
- Limits and uncertainty
- Reproducible practices
- Responsible AI use

Multidisciplinary support teams translate Open Science principles into everyday research practice.

Brazil already has the foundations of a national ecosystem



STRATEGIC ASSETS

- SciELO and open journals
- BDTD and Oasisbr
- BrCris and research integration
- Cariniana and digital preservation
- Data, citizen science, and software networks

The next step is coordinated governance, stable funding, and stronger interoperability.

Ibict can contribute across all six objectives

GOVERNANCE, DATA, INFRASTRUCTURE

- **1 Governance:** policy expertise, networks, standards, and monitoring.
- **2 Open data:** Oasisbr, BDTD, metadata aggregation, and interoperability.
- **3 Infrastructure:** BrCris, Laguna, Cariniana, dARK, and public digital services.

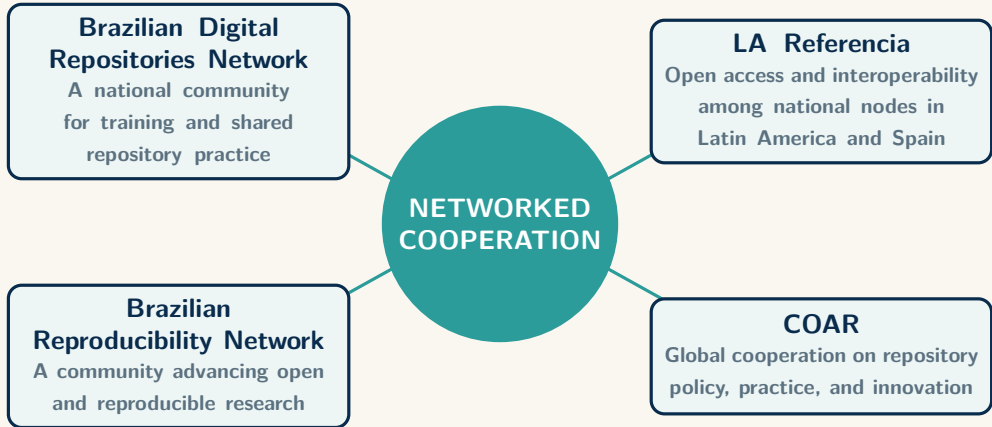
COMMUNICATION, PARTICIPATION, CAPACITY

- **4 Communication:** repositories, journals, open publishing, and persistent access.
- **5 Engagement:** information literacy, citizen science, and multilingual access.
- **6 Capacity:** training, South–South exchange, and open technology transfer.

Brazil can move from endorsing shared principles to delivering interoperable services, professional capacity, and evidence for cooperation.

Mapped to the Action Plan for International Cooperation in Open Science, 2025.

Networks connect national capacity with global cooperation



Networks operationalize Objectives 1 and 3 through shared standards, learning, and collective action.

Ibict connects the national ecosystem in layers

COLLECT AND PUBLISH

- Institutional repositories and digital libraries
- Brazilian Digital Library of Theses and Dissertations
- Journal and publication support services

AGGREGATE AND CONNECT

- Oasisbr for unified discovery
- National repository networks and shared standards
- International exchange through open protocols

ANALYZE AND GOVERN

- BrCris for research information and policy intelligence
- Laguna for semantic integration and large-scale analysis

PRESERVE AND IDENTIFY

- Cariniana for distributed digital preservation
- dARK for persistent, traceable identifiers

Together, these layers preserve institutional autonomy while enabling national-scale discovery and reuse.

Three infrastructures strengthen Brazilian research

BRCRIS Connects the research system

- Researchers and institutions
- Projects and funding
- Outputs and infrastructures

Enables visibility, governance, assessment, and evidence-based policy.

DARK NETWORK Keeps resources identifiable

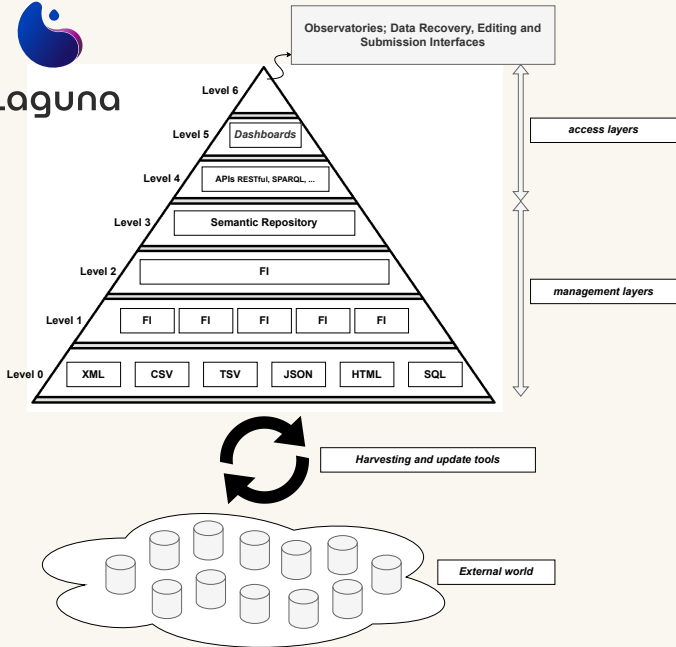
- Persistent ARK identifiers
- Traceability and citation
- Long-term access

Protects reproducibility and institutional memory from broken links.

PROJECT LAGUNA Integrates heterogeneous data

- Semantic enrichment
- Interoperability
- Large-scale discovery

Transforms isolated sources into a shared analytical environment.



BrCris creates a national view of Brazilian research

BRAZILIAN CURRENT RESEARCH INFORMATION SYSTEM

- Developed and led by **Ibict**; officially established in 2014.
- Integrates, retrieves, certifies, and visualizes information about Brazil's research ecosystem.
- Organizes fragmented records into a unified national model aligned with international standards.
- Provides free access for researchers, managers, funders, journalists, and citizens.

15.9M
publications indexed

Platform count displayed in July 2026

A public gateway to understand who produces knowledge, where, with whom, and with what results.

Search in the Brazilian Scientific Research Information Ecosystem (BrCris)

Publications ▼

Enter at least 3 characters and search among 15,910,175 Publications

 Search



BrCris

The Brazilian Scientific Research Information Ecosystem, BrCris, is an aggregator platform that allows retrieving, certifying and visualizing data and

Oasisbr connects more than 6.5 million open digital objects



A NATIONAL AND INTERNATIONAL GATEWAY

6.5M+

articles, theses, dissertations, books, book chapters, and research datasets

~1.2M

documents aggregated through BDTD

~2,000

information sources — repositories and scientific journals across Brazil



BDTD gives visibility to graduate research

A NATIONAL SERVICE SINCE 2002

- Conceived, developed, and maintained by Ibict.
- Integrates theses and dissertations from Brazilian teaching and research institutions.
- Also includes records of works defended by Brazilians at institutions abroad.
- Offers free national discovery; full texts remain hosted by participating institutions.

PUBLIC VALUE

- Visibility for postgraduate research
- Shared metadata standards
- Institutional interoperability
- Access to specialized knowledge
- Accountability for public investment



Oasisbr, BDTD, and BrCris play distinct roles

Service	Primary question	Scope	Strategic role
Oasisbr	What open research outputs can I discover?	Publications, data, books, conference works, theses, and more	Multidisciplinary access and national/international aggregation
BDTD	What graduate research has been produced?	Brazilian theses and dissertations, including records from abroad	Visibility and governance of postgraduate knowledge
BrCris	How is the research ecosystem connected?	Actors, organizations, projects, programs, outputs, patents, and software	Research intelligence, indicators, governance, and policy support

Together they connect **discovery**, **specialized graduate knowledge**, and **system-level intelligence**.

dARK turns persistence into regional sovereignty

WHAT IT ADDS TO ARK

- **Decentralization:** participating institutions share operational responsibility.
- **Traceability:** identifier events remain auditable over time.
- **Resilience:** distributed governance reduces single points of failure.
- **Sovereignty:** regional institutions define policies and priorities.

HOW IT WORKS WITH OASISBR

1. Repositories expose qualified metadata through open protocols.
2. Oasisbr aggregates and validates the records.
3. dARK assigns persistent identifiers to eligible objects.
4. Users discover content centrally while access remains with the source institution.



dARK



National delivery enables international cooperation

NATIONAL BACKBONE

- A focal point linking domestic and international partners
- Defined roles for government, funders, institutions, and networks
- Stable funding for public infrastructures and professional teams
- Shared standards with regional and disciplinary adaptation

MULTILATERAL MECHANISM

- A permanent platform for exchange and joint delivery
- Coordinated resource allocation and practical pilot projects
- Dynamic monitoring and transparent reporting
- Rotating meetings to share evidence and improve practice

Domestic capacity is the foundation for equal participation, joint contribution, and shared benefits.

Measurable commitments make cooperation credible

INFRASTRUCTURE AND KNOWLEDGE

- Platforms connected through open standards
- Data shared with ethical access and benefit safeguards
- Public digital services jointly developed or reused
- Papers, preprints, and multilingual outputs made accessible

PEOPLE AND COOPERATION

- Researchers and professionals trained across borders
- Young people and the public engaged in Open Science
- South–South projects and exchanges sustained
- Technologies transferred and adapted to local needs

Monitor reach, interoperability, equity, reuse, and local capability.

Brazil can help turn the action plan into shared infrastructure



Connect services + launch pilots + train people + share evidence

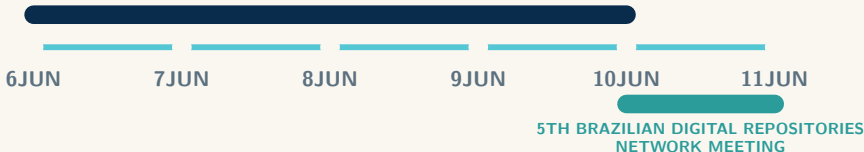
Ibict offers mature public services and regional partnerships that can support equitable, non-commercial, and interoperable cooperation.

Salvador/Brazil connects repository communities in 2027



OpenRepositories

OPEN REPOSITORIES 2027



6–10 JUNE 2027

Open Repositories 2027

Salvador, Bahia — global exchange comes to South America's repository community.

10–11 JUNE 2027

Brazilian Repository Network

The fifth national meeting advances cooperation and shared repository practices.

A 2027 milestone for shared pilots, progress review, and network alignment.



Thank you!

washingtonsegundo@ibict.br

Aguyjevete

Guarani

Ẹ ẹ́

Yoruba

Sulpayki

Quechua

Na gode

Hausa

Asante

Swahili / Bantu

धन्यवाद

Hindi

Ngiyabonga

Zulu / Bantu

ধন্যবাদ

Bengali

நன்றி

Tamil

አመሰግናለሁ

Amharic

Terima
kasih

Indonesian

谢谢

Chinese

Obrigado

Portuguese

